V SEMESTER (PROFESSIONAL ELECTIVE-I)	L	T	P	C
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20AM5E04-DevOps				

COURSE OBJECTIVES:

1. Introduces the basic concepts of Information System.
2. To understand the Management Control Framework and The Application Control Framework.

COURSE OUTCOMES:

At the end of the course, student will be able to

CO1: Enumerate the principles of continuous development and deployment, automation of configuration management, inter-team collaboration, and IT service agility[K3].
CO2: Describe DevOps & DevSecOps methodologies and their key concepts[K2].
CO3: Illustrate the types of version control systems, continuous integration tools, continuous monitoring tools, and cloud models[K2].
CO4: Set up complete private infrastructure using version control systems and CI/CD tools[K2].
CO5: Acquire the knowledge of maturity model, Maturity Assessment[K4].

UNIT - I :

Introduction: Phases of Software Development Life Cycle, Values and princi- ples of agile software development.

UNIT - II :

Fundamentals of DevOps: Architecture, Deployments, Orchestration, Need, Instance of applications, DevOps delivery pipeline, DevOps eco system.

UNIT - III :

DevOps adoption in projects: Technology aspects, Agiling capabilities, Tool stack im- plementation, People aspect, processes

UNIT - IV :

CI/CD: Introduction to Continuous Integration, Continuous Delivery and Deployment, Benefits of CI/CD, Metrics to track CICD practices

UNIT - V :

Devops Maturity Model: Key factors of DevOps maturity model, stages of Devops matu- rity model, DevOps maturity Assessment.

Text Books:

1. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations, Gene Kim , John Willis , Patrick Debois , Jez Humb,1st Edition, O'Reilly publications, 2016.
2. What is Devops? Infrastructure as code, 1st Edition, Mike Loukides O'Reilly publica- tions, 2012.